

April 6, 2026

Silver: A Critical Mineral in the Efficient Transfer of Energy Into Compute

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One important input into my writing is the feedback loop I get from institutional investors: the questions they ask and the responses they give to the AI ideas I raise. I felt it recently with Marvell as most email responses were disagreements. If I find ideas people like, I have not done a good job of trying to find good reward to risk situations. As the hardware trade has developed, investors have found it relatively easy to embrace areas like semiconductors, optical fiber, thermal management, power systems, and many other components critical to AI. One area, however, remains underappreciated even though it may be more critical than all of them in my opinion after much research: silver. The more I bring up silver with investors, the more I find that the dominant reaction is to view it simply as part of the gold trade, which in turn opens up an opportunity.

Plain and simple, silver is a critical mineral in the efficient transfer of energy into compute. Its importance begins with physics. The U.S. Geological Survey notes that pure silver has the highest electrical and thermal conductivity of any metal, which is why it remains important in electrical and electronic products where efficient current flow and effective heat management matter. In the AI era, that is no side detail. As the industry shifts from a training-led buildout toward an inference-led architecture, the economic objective is moving toward lower cost per token, higher throughput per watt, more efficient memory movement, and lower system-level heat and power loss. That is increasingly consistent with Jensen Huang's recent framing around AI factories, token cost, and tokens per watt, particularly in the context of Vera Rubin. Silver matters not because of one isolated use case, but because its physical properties support the broader hardware stack that turns energy into usable computation.

This is the key conceptual shift. In the training phase of AI, investors understandably focused on the obvious bottlenecks: GPUs, high-bandwidth memory, networking, transformers, gas turbines, and massive capex. But inference changes the shape of the problem. The rise of agentic AI is the catalyst, because agents create persistent, multi-step, and often recursive token demand that must be served cheaply, reliably, and continuously. The question is no longer only how to train the biggest model, but how to serve tokens efficiently at scale. That broadens the economics of AI from the chip to the machine, and from the model to the system that supports it. In that world, materials that reduce electrical loss, improve thermal performance, and support system reliability take on greater importance even if they never become the headline component.

The move from training to inference makes heat more important, but the deeper shift is toward electrical and thermal efficiency at the system level. As Blackwell gives way to Rubin, NVIDIA's architecture is increasingly optimized for tokens per watt, tighter cooling loops, and denser inference. In that world, silver's importance rises not simply because systems run hotter, but because power-dense AI infrastructure has less tolerance for electrical loss, thermal inefficiency, and reliability failures.

That is where silver enters the story more directly. Silver is not the processor, but it is one of the materials embedded across the system that helps convert energy into reliable computational output. It shows up in enough of the electrical and thermal architecture to matter: connectors, switches, relays, semiconductor packaging, and other components where electrical conductivity, thermal stability, and heat management affect uptime and performance. The point is not that every component contains silver. It is that silver appears often enough across high-value system functions that scaling AI infrastructure can also scale silver demand.

This is why silver should not be dismissed as just another industrial input. Its role is small by cost but large by function. In a hyperscale AI deployment, the large cost buckets are land, power, transformers, cooling, GPUs, memory, and networking gear. Silver is only a tiny share of the total bill. But that is precisely why its demand can be relatively price-inelastic. If the silver embedded in a relay, connector, or thermal interface material becomes more expensive, the buyer is unlikely to cancel a server deployment or halt a data center buildout. More often, the system absorbs that increase because silver is inexpensive relative to the total value of the hardware it supports. That is one reason silver can matter financially even without dominating the bill of materials.

The inference transition also broadens the silver thesis beyond the cloud. Edge AI pushes the same logic further. At the edge, power budgets, thermal limits, and spatial constraints are even tighter than in a hyperscale campus. A giant data center can throw more capital, more cooling equipment, and more electrical engineering at a bottleneck. An edge device has much less room for inefficiency. As AI moves outward into enterprise hardware, industrial devices, robotics, autonomous systems, and edge inference, the premium on efficient power transfer, reliable electrical connections, and effective heat management rises. Silver is not the singular enabler of that world, but it is one of the enabling minerals that benefits when the entire stack is forced to become more electrically efficient, more thermally stable, and more operationally reliable. The more AI shifts from a brute-force compute race to an efficiency race, the stronger the strategic case for materials that help reduce losses.

This is happening against a backdrop of already tightening industrial demand. The Silver Institute reported that industrial silver demand rose 4% in 2024 to a record 680.5 million ounces, the fourth consecutive annual record, driven by grid expansion, electrification, photovoltaic deployment, and AI-related applications. It then projected in early 2025 that industrial demand would rise again and exceed 700 million ounces for the first time. That is important because AI is not creating the silver story from scratch. It is arriving on top of an already tight industrial market in which silver was already being pulled by electrification and energy-transition demand. The result is a more powerful cumulative case: silver is increasingly leveraged to several secular growth drivers at once, not just one.

Solar reinforces the same economic lesson. In photovoltaics, silver use per cell has declined over time as manufacturers have worked to reduce loadings and improve process efficiency. Yet total silver demand has remained substantial because deployment has grown so quickly. This is the right framework for AI as well. It is entirely possible that engineers will continue trying to lower silver usage per component where possible. But the more important question is whether system-level scale overwhelms per-unit thrift. If the world is building far more servers, switches, campuses, power systems, and edge devices, then aggregate demand can still rise even if each unit uses silver more efficiently. Silver's importance comes not from one spectacular loading number, but from repeated use across a rapidly expanding installed base. Given the rise in oil prices due to the war and the clear need for domestic energy independence amid Strait of Hormuz risk, solar has to be part of the answer for the world. Meanwhile, silver has cheapened relative to oil, and if you convert that relationship into BTUs, the decline is even larger.

The same logic extends beyond civilian infrastructure into defense. Silver was added to the final 2025 U.S. critical minerals list. USGS explains that minerals are deemed critical when they are essential to the economy and to national security and when their supply chains are vulnerable to disruption. That designation matters for the paper because it validates the broader claim that silver is no longer just an overlooked precious metal. If you look for Gold on the list, you won't find it which only reinforces the point. Silver has become strategically relevant enough to be recognized in official U.S. critical-minerals policy. Modern defense systems are increasingly electronic, power-dense, and reliability-sensitive, relying on sensors, communications systems, ruggedized connectors, batteries, control modules, and thermal-management systems. Those platforms benefit from the same material properties that make silver useful in AI infrastructure: conductivity, thermal performance, and reliability in demanding conditions.

Using a drone-versus-tank example is useful not because it gives us a precise bill of materials, but because it reveals the economic logic of silver in modern warfare. There are no reliable public disclosures showing exactly how much silver sits inside a \$5,000 drone or a tank worth well over \$1 million. But the broader conclusion is still clear. A low-cost drone likely contains only a small amount of silver spread across electronics, connectors, power-management components, and switching systems. A tank almost certainly contains materially more silver in aggregate because it carries far more wiring, sensors, radios, and ruggedized electronic subsystems. Yet in both cases, silver remains a tiny share of total platform cost while serving functions that are essential to performance and reliability. That is the real point. Silver is not economically

important because it is expensive. It is economically important because it is cheap, functionally critical, and therefore very difficult to economize away when the mission matters.

The numbers in Ukraine make the multiplication effect clear. Reuters reported that Ukraine planned to purchase about 4.5 million FPV drones in 2025, after buying more than 1.5 million in 2024, while President Zelenskiy said the country had the capacity to produce 4 million drones annually. By contrast, the International Institute for Strategic Studies estimated that Russia lost about 1,400 main battle tanks in 2024. The exact silver content of a military FPV drone is not publicly disclosed, and that uncertainty should be acknowledged. But once warfare shifts toward millions of low-cost, expendable drone units per year, even very small per-unit amounts of silver embedded in electronics, switching systems, and control modules can become meaningful in aggregate. The same multiplication logic applies in AI. No single server, switch, power distribution unit, connector set, or thermal-control layer needs to contain a dramatic amount of silver for the demand story to matter. What matters is replication. Hyperscale buildouts spread small amounts of silver across entire campuses, just as drone warfare spreads small material inputs across enormous fleets. In both cases, silver benefits from the same underlying dynamic: a low-cost but high-function material embedded across platforms now being produced, deployed, and scaled.

A useful mental anchor for scale comes from electronics recycling. The EPA says that recycling one million cell phones can recover 772 pounds of silver, which works out to roughly 0.35 grams per phone. A military drone is obviously not a phone, and a server or power unit is not either, so that figure should not be transferred mechanically. But it does illustrate the core principle: very small amounts of silver can still add up to meaningful totals when multiplied across millions of electronic devices. That is the heart of the silver case in both AI and defense. Per-unit usage may look trivial, but once those units are deployed at scale and silver sits at functionally critical points, aggregate demand can become significant.

There is also a supply-side reason this matters. The World Silver Survey 2025 reported a structural silver market deficit of 148.9 million ounces in 2024, the fourth consecutive annual deficit, and cumulative deficits of 678.4 million ounces over 2021–2024. Crucially, this deficit is compounded by the fact that silver supply is uniquely inelastic to its own price movements. Roughly 80% of the world's silver is mined as a byproduct of extracting base metals like lead, zinc, and copper. Even if AI demand and electrification demand push silver prices significantly higher, miners cannot simply turn on more silver production; their output is fundamentally dictated by the production economics of those primary base metals. The market cannot easily drill its way out of a silver shortage.

The same Survey and Silver Institute commentary indicate that industrial demand has been a key driver of the tight backdrop. That does not guarantee a straight-line price move, and it would be too strong to argue that every bit of AI infrastructure automatically translates into immediate shortages. But it does mean the market is starting from a tighter position than casual observers often assume. If AI infrastructure, defense systems, solar deployment, and electrification investment all keep pulling on the same mineral, then silver's strategic value can rise even before the market fully appreciates where the bottlenecks are.

The deeper investment point is that silver's role becomes more important as AI becomes more physical. The early years of the AI boom were dominated by software fascination and compute scarcity. The next phase looks more like industrial electrification: token factories, racks optimized for efficiency, whole-campus power design, edge deployment, and the conversion of electricity into persistent inference. NVIDIA's own language increasingly reflects that shift toward performance per watt and revenue per megawatt. Once AI is seen through that lens, silver becomes easier to understand. It is not just a precious metal with a speculative following. It is a critical mineral embedded in the systems that make efficient compute possible. That is a much stronger and more durable framing.

This is why silver's importance should be described in functional, not decorative, terms. Copper gets most of the attention when investors think about electrification because it is the bulk metal of the grid, the cables, the motors, and the large-scale power systems. But embodied AI changes that framing. As compute moves into robots, drones, autonomous systems, edge devices, and power-dense inference hardware, the system needs not just more electricity, but better electrical performance under tighter thermal limits and stricter reliability demands. That is where silver matters. It helps modern systems transfer electricity with lower loss, manage heat under heavy load, and maintain reliable operation across dense electronic architectures. That is valuable in data centers. It is valuable in edge devices. It is valuable in solar. It is valuable in modern warfare. And because silver is usually a tiny share of total system cost, demand for those functions can remain resilient even if the metal itself becomes more expensive.

This paper began with a simple observation from my conversations with investors: most still do not think of silver as a critical mineral and instead largely view it as just another precious metal. But that gap between perception and reality is precisely where the opportunity lies. Silver plays a critical role in the efficient transfer of energy into compute. The shift from training to inference only strengthens that case, because the industry is increasingly focused on tokens per watt, lower cost per token, and whole-system electrical and thermal efficiency. Copper remains the volume metal of electrification, but silver is increasingly the performance metal of embodied AI. That matters even more because silver is not entering this demand wave from a position of abundance. The market has already been running persistent structural deficits, meaning AI infrastructure, defense systems, solar deployment, and electrification investment are all pulling on an already tight material base. As AI infrastructure expands from giant clusters to campuses, enterprises, edge devices, and physical systems, silver looks less like an overlooked industrial input and more like a strategic mineral for a hardware-intensive future. It is not the chip. It is not the rack. But it sits inside enough of the system that, when the whole system matters more, silver matters more. Silver has been in a multi-year structural deficit, while we enter the next phase of AI around agentic systems, physical AI, and performance-per-watt economics.

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